Project State Report

Unified Multi-Domain Industrial Inspection Platform

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Architecture, post-processing logic, and model-performance analysis for Stator, Chignon, and File domains.

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1 Executive Summary

The application now runs as a **unified multi-domain inference system** with independent runtime paths for:

- **Stator** detection,
- **Chignon** detection,
- **File** detection.

The infrastructure for all domains is integrated in one Flask + WebRTC application. Model behavior and post-processing are domain-specific.

Critical model decision (File domain): YOLO file-domain runs are non-convergent in current training outputs (mask mAP metrics collapse to 0.0), so deployment uses **UNet-ResNet18 (.pth)** for file detection.

2 Application Architecture

2.1 Domain-Aware Runtime

Requests are routed by domain to dedicated managers:

- Stator → `ModelManager`
- Chignon → `ChignonModelManager`
- File → `FileModelManager`

This routing is applied consistently to model listing, model switching, enhancement settings, upload detection, and WebRTC live inference.

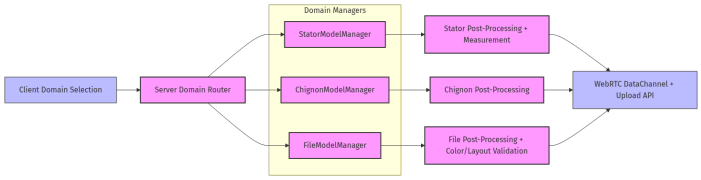


Figure 1: Multi-domain architecture: one app, three domain managers, domain-specific post-processing.

This architecture shows the core orchestration principle of the application: a single UI and transport layer delegates inference to independent domain managers. Each manager owns model discovery, model loading, prediction, and post-processing semantics for its domain. This separation avoids cross-domain rule leakage (for example, stator geometry rules affecting file logic) while preserving one deployment surface for APIs, WebRTC streaming, and settings.

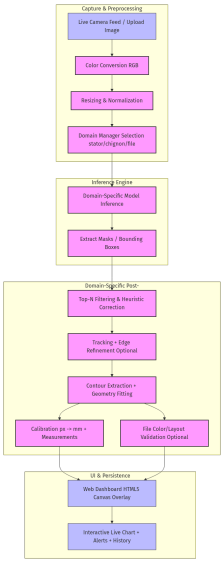


Figure 2: Unified detection flow with domain manager selection stage.

The unified flow emphasizes that preprocessing remains shared, but decision logic diverges right after domain selection. Input frames are normalized once, then routed to the selected manager where detection outputs are transformed by domain-specific post-processing. This design keeps latency predictable and allows independent evolution of stator measurement, chignon segmentation tuning, and file color-layout validation without changing the transport pipeline.

2.2 Separate Detection Pipelines

Each domain has its own processing flow due to different industrial constraints.

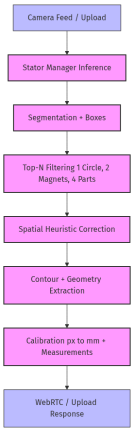


Figure 3: Stator pipeline: top-N filtering, spatial correction, contour geometry, and measurement calibration.

The stator pipeline is measurement-centric. After raw detections, top-N constraints enforce physically valid object counts, then a spatial heuristic corrects class confusion using position relative to the stator center. Contours are converted to geometric primitives, and calibrated scale factors convert pixel distances into millimeter-level outputs for QA metrics such as width, diameter, area, and inter-part spacing.

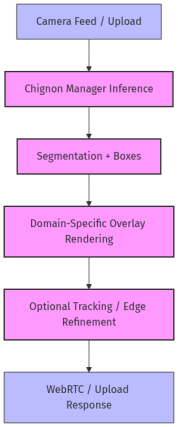


Figure 4: Chignon pipeline: domain-specific segmentation and rendering with optional tracking/edge refinement hooks.

The chignon pipeline prioritizes stable segmentation and visualization fidelity over heavy geometric measurement. Domain-specific rendering logic controls class overlays and confidence presentation, while optional tracking/edge-refinement hooks can be enabled to smooth temporal behavior and contour boundaries. This keeps the path lightweight and suited for real-time monitoring with minimal rule-based overrides.

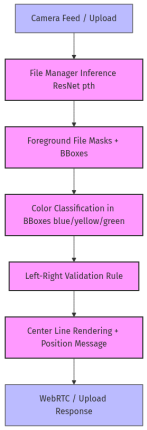


Figure 5: File pipeline: ResNet inference, color classification, left-right validation, and correctness feedback.

The file pipeline is validation-centric and currently anchored on UNet-ResNet18 outputs. Detected file regions are post-processed by color scoring inside bounding boxes to classify blue/yellow semantics, then a left-right rule checks assembly positioning. The system draws a center line between paired files and emits explicit correctness messages to the UI, enabling operators to interpret not only detection presence but also placement conformity.

3 Model Results Analysis by Domain

3.1 Stator Domain Results

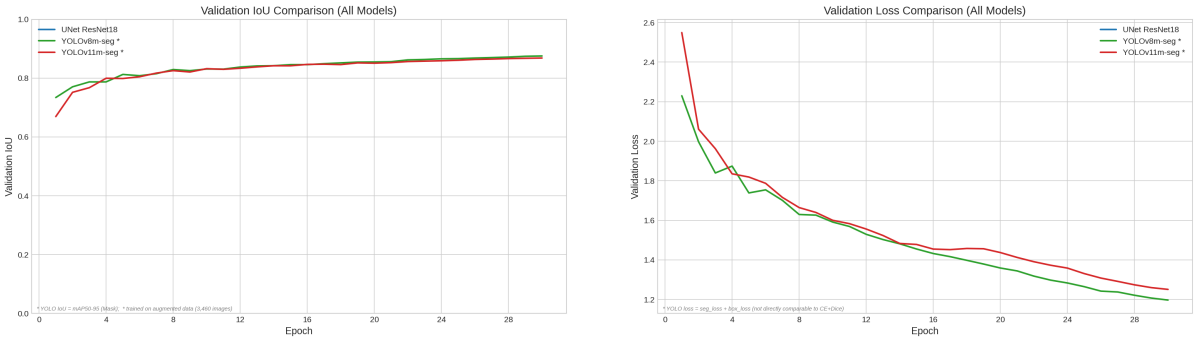
Training artifacts analyzed from `outputs/results`.

Model	Best Epoch	Primary Metric	Value	Peak GPU (MB)
UNet-ResNet18	30	Val IoU	0.9663	472.5
YOLOv8m-seg	30	Mask mAP50-95	0.8751	–
YOLOv11m-seg	30	Mask mAP50-95	0.8682	–

Table 1: Stator model summary (from training logs and YOLO results CSV).

Interpretation:

- UNet-ResNet18 provides strongest pixel-level IoU for stator segmentation.
- YOLO models maintain high precision/recall and strong instance-level mask mAP.
- Stator domain supports both YOLO and PyTorch runtime paths in production.



(a) Stator IoU curves

(b) Stator loss curves

Figure 6: Stator training dynamics.

3.2 Chignon Domain Results

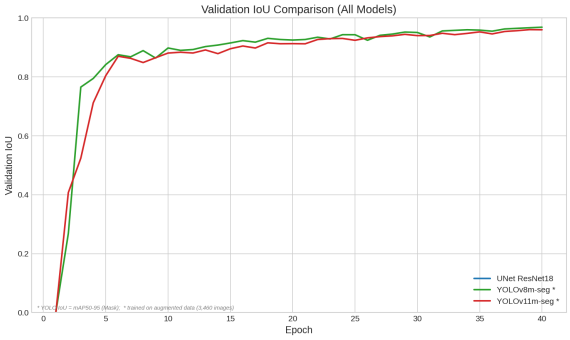
Training artifacts analyzed from `chignon/results`.

Model	Best Epoch	Primary Metric	Value	Peak GPU (MB)
UNet-ResNet18	39	Val IoU	0.9816	440.9
YOLOv8m-seg	40	Mask mAP50-95	0.9688	–
YOLOv11m-seg	39	Mask mAP50-95	0.9606	–

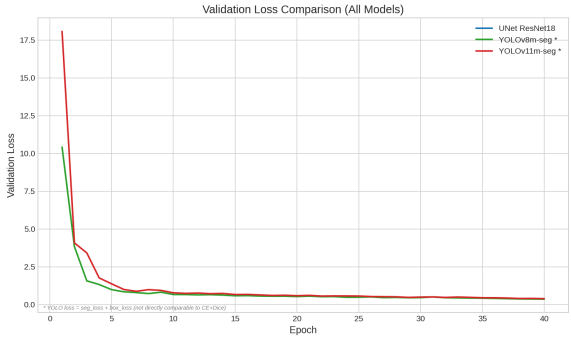
Table 2: Chignon model summary (from training logs and YOLO results CSV).

Interpretation:

- Chignon domain converges strongly for both UNet and YOLO variants.
- YOLO recall is saturated at 1.0 in best checkpoints, indicating high object recovery on this dataset.
- Domain is integrated in app routing and remains suitable for YOLO/PyTorch comparison workflows.



(a) Chignon IoU curves



(b) Chignon loss curves

Figure 7: Chignon training dynamics.

3.3 File Domain Results

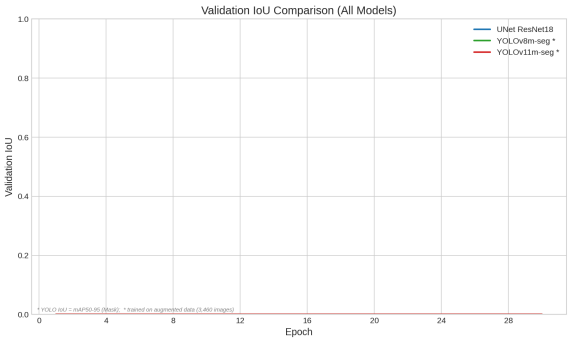
Training artifacts analyzed from `files/results`.

Model	Best Epoch	Primary Metric	Value	Peak GPU (MB)
UNet-ResNet18	30	Val IoU	0.9197	440.9
YOLOv8m-seg	1	Mask mAP50-95	0.0000	–
YOLOv11m-seg	1	Mask mAP50-95	0.0000	–

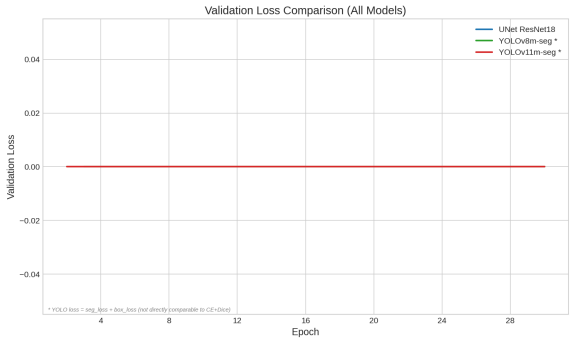
Table 3: File model summary (from training logs and YOLO results CSV).

Interpretation:

- File-domain YOLO training failed to produce deployable mask quality (all key mask metrics collapse to zero).
- UNet-ResNet18 remains the only viable model path for file-domain inference in this cycle.
- Deployment therefore prioritizes UNet-ResNet18 plus file-specific post-processing rules.



(a) File IoU curves



(b) File loss curves

Figure 8: File-domain training dynamics.

4 Post-Processing Methods by Domain

4.1 Stator Post-Processing

- Top-N filtering limits structurally valid object counts.
- Spatial heuristic correction mitigates class confusion based on angular position around stator center.
- Contour extraction + geometry fitting computes dimensions.
- Calibration methods (manual/reference/intrinsics/depth) map px to mm.

4.2 Chignon Post-Processing

- Domain-specific segmentation rendering and object annotation.
- Optional enhancement hooks for tracking and edge refinement.
- Lightweight geometric post steps compared to stator measurement-heavy path.

4.3 File Post-Processing

- Per-bounding-box color analysis (blue/yellow/green scoring).
- Left-right layout rule validation with correctness message.
- Center-line drawing between selected file pair.
- UI toggle to enable/disable file color-layout validation at runtime.

5 Deployment Decision and Current Status

Domain	Active Runtime Strategy	State
Stator	YOLO + UNet with measurement pipeline	Stable
Chignon	YOLO + UNet via dedicated manager	Integrated
File	UNet-ResNet18 + color/layout validation	Integrated, model lifecycle experimental

Table 4: Current deployment state by domain.

6 Conclusion

The report confirms the platform has matured into a domain-aware architecture with separate detection pipelines and post-processing methods. Stator and Chignon domains show strong model convergence, while File-domain YOLO training is currently non-viable. As a result, file deployment uses UNet-ResNet18 with dedicated color/layout validation logic until a successful YOLO retraining cycle is achieved.